# William Taylor

Computer Architecture DPhil Student, Department of Engineering Science, University of Oxford
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## Profile

Computer Architecture DPhil student at the University of Oxford researching next-generation memory architectures and memory optimisations for more efficient AI systems. Member of the Oxford Secure Computer Architecture Research (OSCAR) group and contributor to the Computing Infrastructure Group. DPhil fully funded by the Advanced Research and Invention Agency (ARIA) through Project GLASS, which aims to design the next generation of AI systems from the physical layer to the application layer.
**Research interests:** Computer Architecture · Memory Systems · Hardware-based Security · Accelerators · Parallelism

## Education

**DPhil (PhD) in Engineering Science** Oct 2025 – present
*University of Oxford* *Oxford, UK*

- Supervisors: Prof. Amro Awad and Prof. Noa Zilberman
- Research: data-aware, bandwidth-efficient memory architectures for GPU-based AI accelerators
- Fully funded by ARIA as part of Project GLASS (Glass-based Scalable AI System)

**MEng Electrical and Electronic Engineering** 2021 – 2025
*University of Nottingham* *Nottingham, UK*

- First Class Honours
- MEng thesis: *Designing FPGAs for Cube Satellites to Detect and Monitor GNSS Interference* (Vivado, ModelSim, Skydel GNSS Simulator, MATLAB spectrum analysis)

## Publications

**William Taylor**, Sunho Lee, Henry Wang, Noa Zilberman, and Amro Awad. "VERVE: Verifiable Value Retrieval Leveraging Value Reuse in Accelerator Memory." In *Proceedings of the 44th IEEE International Conference on Computer Design (ICCD)*, November 2026.

## Research and Industry Experience

**Research Assistant** Aug 2025 – Sep 2025
*University of Oxford, Department of Engineering Science* *Oxford, UK*

- Researched GPU memory architectures for AI acceleration under Prof. Amro Awad
- Built and ran GPU simulation workflows with Accel-Sim/GPGPU-Sim and PyTorch on Linux

**Hardware Systems Intern (R&D Lab)** Jun 2024 – Sep 2024
*Qualcomm Technologies* *Cambridge, UK*

- Testing and test automation of wireless and audio boards, mobile devices and test instrumentation
- Developed coding frameworks using Python, SCPI and GitHub, working both in a team and independently

**Chip Implementation Intern (Physical Design and Synthesis)** Jun 2023 – Sep 2023
*Qualcomm Technologies* *Cambridge, UK*

- Implemented chip designs using Qualcomm mixed-signal IP for upcoming tape-outs, focusing on synthesis and physical design
- Worked with Verilog, TCL/Perl and Cadence and Synopsys EDA tools; design management with ClearCase and DesignSync

## Awards, Service and Leadership

- **ARIA-funded DPhil studentship**, Project GLASS, University of Oxford (2025–present)
- **UK Electronics Skills Foundation (UKESF) Scholar**, sponsored by Qualcomm (2023–2025); volunteered at conferences and mentored other UKESF scholars
- **President**, Electrical and Electronic Engineering Society, University of Nottingham

## References

**Prof. Amro Awad**
Department of Engineering Science
University of Oxford
*Contact details on request*

**Prof. Noa Zilberman**
Department of Engineering Science
University of Oxford
*Contact details on request*